

Internal Audit Certificate

Paper III — Linear-Error Clique Partitions of Split Graphs (Erdos #81)

Audit block: Block 02 — Common-profile LP: $\nu_3^*(H(p,q,d)) = F(p,q,d)$ (Thm 3.1)

VERDICT: PASS (LP 351/351, EXACT 351/351)

Scope

Direct fractional triangle-packing LP on the actual graph $H(p,q,d)$ (SciPy HiGHS) compared against the closed form $F(p,q,d)$, over the grid $3 \leq p \leq 8$, $0 \leq q \leq 8$, $0 \leq d \leq p$.

Summary of evidence

- (LP) 351/351: direct-graph HiGHS packing LP optimum equals closed form F .
- (EXACT) 351/351: exact-rational feasible-cover certificate $\nu_3^* \leq F$.
- Maximum $|LP - F|$ over the grid = $3.9e-14$; exact cover removes float reliance.

Methodology

All numeric checks use exact rational arithmetic (Python fractions) or exact linear/integer programming; symbolic identities are proved with SymPy ($\text{simplify}(\text{LHS}-\text{RHS})=0$ or sum-of-squares certificate). The audit re-derives each claim independently of the manuscript's own scripts and of the Lean formalization.

Reproducibility

Results file: results/common_profile_LP_results.txt

SHA-256 (results file): 1e977ecba3d6475adf02330f8fb6d0c382d6123d83ead0bb468c7287c3e79773

Generated: 2026-07-21 19:38 UTC | Tooling: Python 3.14, SymPy 1.14, SciPy 1.17, PuLP/CBC

Auditor: internal automated audit harness (Claude Code)

This certificate attests to computational/symbolic verification of the stated finite/closed-form claims at the audited parameter ranges. It complements, and is independent of, the machine-checked Lean 4 / Mathlib formalization.